# Answers learning task 3

2026 – 2027

For exercises 3.1 and 3.5, determine the range (possible values) of the random variable.

**Exercise 3.1:** The random variable is the number of nonconforming solder connections on a printed circuit board with $2000$ connections.

Answer

The range of this discrete random variable is the set $\{0, 1, 2, \ldots, 2000\}$, as each of the $2000$ connections can be nonconforming or not.

**Exercise 3.5:** A batch of $600$ machined parts contains $10$ that do not conform to customer requirements. Parts are selected successively, without replacement, until a nonconforming part is obtained. The random variable is the number of parts selected.

Answer

Notice that we keep picking until a nonconforming part is obtained, so the lower bound is $1$ if we immediately pick one, and can take up to $591$ tries if we have bad luck and first take all $590$ conforming parts before we pick a nonconforming one. Therefore, the range of this discrete random variable is the set $\{1, 2, 3, \ldots, 591\}$.

**Exercise 3.14:** The sample space of a random experiment is $\{a, b, c, d, e, f\}$, and each outcome is equally likely. A random variable is defined as follows:

| **Outcome** | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| $x$ | 0 | 0 | 1.5 | 1.5 | 2 | 3 |

Determine the probability mass function of $X$. Use the probability mass function to determine the following probabilities:

(a) $P(X = 2)$
(b) $P(0.6 < X < 2.7)$
(c) $P(X > 3)$
(d) $P(0 < X \leq 2)$
(e) $P(X = 0 \text{ or } X = 2)$

Answer

Each of the outcomes of the random experiment occurs with equal probability, so we can use the classical approach to probability theory:

(a) $P(X = 2) = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}} = \frac{1}{6} \approx 0.1667$,

(b) $P(0.6 < X < 2.7) = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}} = \frac{3}{6} = 0.5$,

(c) $P(0.6 < X < 2.7) = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}} = \frac{3}{6} = 0.5$,

(d) $P(0 < X \leq 2) = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}} = \frac{5}{6} \approx 0.8333$,

(e) $P(X = 0 \text{ or } X = 2) = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}} = \frac{3}{6} = 0.5$.

**Exercise 3.17:** Verify that the following function is a probability mass function, and determine the requested probabilities.

$$f(x) = \frac{2x+1}{25}, \qquad x = 0, 1, 2, 3, 4$$

(a) $P(X = 3)$
(b) $P(X \leq 1)$
(c) $P(2 \leq X < 4)$
(d) $P(X > -10)$

Answer

This function is a probability mass function, since for each possible value of $x$ the function value $f(x) \geq 0$. Furthermore, the sum for $x = 0, 1, 2, 3, 4$ of $f(x)$ equals $1$.

(a) $P(X = 3) = f(3) = \frac{2 \cdot 3 + 1}{25} = \frac{7}{25} = 0.28$.

(b) $P(X \leq 1) = f(0) + f(1) = \frac{2 \cdot 0 + 1}{25} + \frac{2 \cdot 1 + 1}{25} = \frac{1}{25} + \frac{3}{25} = 0.16$.

(c) $P(2 \leq X < 4) = f(2) + f(3) = \frac{2 \cdot 2 + 1}{25} + \frac{2 \cdot 3 + 1}{25} = \frac{5}{25} + \frac{7}{25} = 0.48$.

(d) All the probability mass lies on outcomes $x = 0, 1, 2, 3, 4$, which are larger than $10$. Therefore: $P(X > -10) = f(0) + f(1) + f(2) + f(3) + f(4) = 1$.

**Exercise 3.29:** An assembly consists of three mechanical components. Suppose that the probabilities that the first, second, and third components meet specifications are $0.95$, $0.98$, and $0.97$, respectively. Assume that the components are independent. Determine the probability mass function of the number of components in the assembly that meet specifications.

Answer

Define the events $M_1$, $M_2$ and $M_3$ indicating whether the three components meet specifications. Then $P(M_1) = 0.95$, $P(M_2) = 0.98$ and $P(M_3) = 0.97$. Furthermore, the events are independent. Then, the number of components meeting specifications has the following probability mass function:

- $k = 0$: neither of the three events occur, so $f(0) = P(M_1' \cap M_2' \cap M_3') = P(M_1') \cdot P(M_2') \cdot P(M_3') = 0.05 \cdot 0.02 \cdot 0.03 = 0.00003$.
- $k = 1$: exactly one of the three events occur:

$$\begin{aligned} f(1) &= P(M_1 \cap M_2' \cap M_3') + P(M_1' \cap M_2 \cap M_3') + P(M_1' \cap M_2' \cap M_3) \\ &= 0.95 \cdot 0.02 \cdot 0.03 + 0.05 \cdot 0.98 \cdot 0.03 + 0.05 \cdot 0.02 \cdot 0.97 \\ &= 0.00301. \end{aligned}$$

- $k = 2$: exactly two of the three events occur:

$$\begin{aligned} f(2) &= P(M_1 \cap M_2 \cap M_3') + P(M_1 \cap M_2' \cap M_3) + P(M_1' \cap M_2 \cap M_3) \\ &= 0.95 \cdot 0.98 \cdot 0.03 + 0.95 \cdot 0.02 \cdot 0.97 + 0.05 \cdot 0.98 \cdot 0.97 \\ &= 0.09389. \end{aligned}$$

- $k = 3$: all three events occur:

$$f(3) = P(M_1 \cap M_2 \cap M_3) = 0.95 \cdot 0.98 \cdot 0.97 = 0.90307.$$

Summarized in a table, we have:

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $f(x)$ | 0.00003 | 0.00301 | 0.09389 | 0.90307 |

**Exercise 3.35:** Determine the cumulative distribution function of the random variable with the following probability mass function:

| $x$ | $-2$ | $-1$ | 0 | 1 | 2 |
|---|---|---|---|---|---|
| $f(x)$ | $1/8$ | $2/8$ | $2/8$ | $2/8$ | $1/8$ |

Furthermore, determine the following probabilities:

(a) $P(X \leq 1.25)$
(b) $P(X \leq 2.2)$
(c) $P(-1.1 < X \leq 1)$
(d) $P(X > 0)$

Answer

We can determine the cumulative distribution function (CDF) $F(x)$ by adding probabilities from left to right:

| $x$ | $-2$ | $-1$ | 0 | 1 | 2 |
|---|---|---|---|---|---|
| $f(x)$ | $1/8$ | $2/8$ | $2/8$ | $2/8$ | $1/8$ |
| $F(x)$ | $1/8$ | $3/8$ | $5/8$ | $7/8$ | 1 |

From this table, we can read the following probabilities:

1. $P(X \leq 1.25) = P(X \leq 1) = \frac{7}{8} = 0.875$,
2. $P(X \leq 2.2) = P(X \leq 2) = 1$,
3. $P(-1.1 < X \leq 1) = P(X \leq 1) - P(X \leq -1.1) = \frac{7}{8} - \frac{1}{8} = 0.75$,
4. $P(X > 0) = 1 - P(X \leq 0) = 1 - \frac{5}{8} = 0.375$.

**Exercise 3.43:** Verify that the following function is a cumulative distribution function, and determine the probability mass function and the requested probabilities:

$$F(x) = \begin{cases} 0 & x < -10 \\ 0.3 & -10 \leq x < 30 \\ 0.7 & 30 \leq x < 50 \\ 1 & 50 \leq x \end{cases}$$

(a) $P(X \leq 50)$
(b) $P(X \leq 40)$
(c) $P(40 \leq X \leq 60)$
(d) $P(X < 0)$
(e) $P(0 \leq X < 10)$
(f) $P(-10 < X < 10)$

Answer

We see that the function $F(x)$ is $0$ for small $x$ and $1$ for large $x$, and it is monotonic non-decreasing (as going to the right we add more and more probabilities). The probability mass function is given by taking the boundary values for $x$ for distinct cases and take the difference in $F(x)$:

| $x$ | $-10$ | $30$ | $50$ |
|---|---|---|---|
| $f(x)$ | $0.3$ | $0.4$ | $0.3$ |

From this, we can derive the probabilities

1. $P(X \leq 50) = F(50) = 1$,
2. $P(X \leq 40) = P(X \leq 30) = F(30) = 0.7$,
3. $P(40 \leq X \leq 60) = P(X = 50) = f(50) = 0.3$,
4. $P(X < 0) = P(X = -10) = f(-10) = 0.3$,
5. $P(0 \leq X < 10) = 0$,
6. $P(-10 < X < 10) = 0$.

**Exercise 3.51:** Determine the mean and variance of the following random variable:

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $f(x)$ | $1/8$ | $2/8$ | $2/8$ | $2/8$ | $1/8$ |

Answer

We can determine the mean of the random variable $X$ as follows:

$$\begin{aligned} E(X) &= \sum_{x=-2}^{2} x \cdot f(x) \\ &= -2 \cdot 1/8 + -1 \cdot 2/8 + 0 \cdot 2/8 + 1 \cdot 2/8 + 2 \cdot 1/8 \\ &= 0. \end{aligned}$$

We can determine the variance of the random variable $X$ as follows:

$$\begin{aligned} \text{Var}(X) &= \sum_{x=-2}^{2} (x - E(X))^2 \cdot f(x) \\ &= (-2-0)^2 \cdot 1/8 + (-1-0)^2 \cdot 2/8 + (0-0)^2 \cdot 2/8 + (1-0)^2 \cdot 2/8 + (2-0)^2 \cdot 1/8 \\ &= 1.5. \end{aligned}$$

**Exercise 3.53:** Determine the mean and variance of the following random variable with the following probability mass function:

$$f(x) = \frac{2x+1}{25}, \qquad x = 0, 1, 2, 3, 4$$

Answer

We can determine the mean of the random variable $X$ as follows:

$$\begin{aligned} E(X) &= \sum_{x=0}^{4} x \cdot f(x) \\ &= 0 \cdot 1/25 + 1 \cdot 3/25 + 2 \cdot 5/25 + 3 \cdot 7/25 + 4 \cdot 9/25 \\ &= 2.8. \end{aligned}$$

We can determine the variance of the random variable $X$ as follows:

$$\begin{aligned} \text{Var}(X) &= \sum_{x=0}^{4} (x - E(X))^2 \cdot f(x) \\ &= (0 - 2.8)^2 \cdot 1/25 + (1 - 2.8)^2 \cdot 3/25 + (2 - 2.8)^2 \cdot 5/25 + (3 - 2.8)^2 \cdot 7/25 + (4 - 2.8)^2 \cdot 9/25 \\ &= 1.36. \end{aligned}$$

**Exercise 3.67:** Let the random variable $X$ have a discrete uniform distribution on the integers $1 \leq x \leq 5$. Determine the mean and variance of $X$.

Answer

The discrete uniform distribution on the integers $1 \leq x \leq 5$ has the following probability mass function:

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $f(x)$ | $1/5$ | $1/5$ | $1/5$ | $1/5$ | $1/5$ |

We can determine the mean of the random variable $X$ as follows:

$$\begin{aligned} E(X) &= \sum_{x=1}^{5} x \cdot f(x) \\ &= 1 \cdot 1/5 + 2 \cdot 1/5 + 3 \cdot 1/5 + 4 \cdot 1/5 + 5 \cdot 1/5 \\ &= 3. \end{aligned}$$

We can determine the variance of the random variable $X$ as follows:

$$\begin{aligned} \text{Var}(X) &= \sum_{x=1}^{5} (x - E(X))^2 \cdot f(x) \\ &= (1 - 3)^2 \cdot 1/5 + (2 - 3)^2 \cdot 1/5 + (3 - 3)^2 \cdot 1/5 + (4 - 3)^2 \cdot 1/5 + (5 - 3)^2 \cdot 1/5 \\ &= 2. \end{aligned}$$

**Exercise 3.68:** Thickness measurements of a coating process are made to the nearest hundredth of a millimeter. The thickness measurements are uniformly distributed with values $0.14$, $0.15$, $0.16$, $0.17$, $0.18$, $0.19$, and $0.20$. Determine the mean and variance of the coating thickness for this process.

Answer

The discrete uniform distribution on these values has the following probability mass function:

| $x$ | 0.14 | 0.15 | 0.16 | 0.17 | 0.18 | 0.19 | 0.20 |
|---|---|---|---|---|---|---|---|
| $f(x)$ | $1/7$ | $1/7$ | $1/7$ | $1/7$ | $1/7$ | $1/7$ | $1/7$ |

We can determine the mean of the random variable $X$ as follows:

$$\begin{aligned} E(X) &= \sum_x x \cdot f(x) \\ &= 0.14 \cdot 1/7 + 0.15 \cdot 1/7 + \ldots + 0.20 \cdot 1/7 \\ &= 0.17. \end{aligned}$$

We can determine the variance of the random variable $X$ as follows:

$$\begin{aligned} \text{Var}(X) &= \sum_x (x - E(X))^2 \cdot f(x) \\ &= (0.14 - 0.17)^2 \cdot 1/7 + (0.15 - 0.17)^2 \cdot 1/7 + \ldots + (0.20 - 0.17)^2 \cdot 1/7 \\ &= 0.0004. \end{aligned}$$

**Exercise 3.70:** The lengths of plate glass parts are measured to the nearest tenth of a millimeter. The lengths are uniformly distributed with values at every tenth of a millimeter starting at $640.0$ and continuing through $640.9$. Determine the mean and variance of the lengths.

Answer

The discrete uniform distribution on these values has the following probability mass function:

| $x$ | 640.0 | 640.1 | 640.2 | 640.3 | 640.4 | 640.5 | 640.6 | 640.7 | 640.8 | 640.9 |
|---|---|---|---|---|---|---|---|---|---|---|
| $f(x)$ | $1/10$ | $1/10$ | $1/10$ | $1/10$ | $1/10$ | $1/10$ | $1/10$ | $1/10$ | $1/10$ | $1/10$ |

We can determine the mean of the random variable $X$ as follows:

$$\begin{aligned} E(X) &= \sum_x x \cdot f(x) \\ &= 640.0 \cdot 1/10 + 640.1 \cdot 1/10 + \ldots + 640.9 \cdot 1/10 \\ &= 640.45. \end{aligned}$$

We can determine the variance of the random variable $X$ as follows:

$$\begin{aligned} \text{Var}(X) &= \sum_x (x - E(X))^2 \cdot f(x) \\ &= (640.0 - 640.45)^2 \cdot 1/10 + (640.1 - 640.45)^2 \cdot 1/10 + \ldots + (640.9 - 640.45)^2 \cdot 1/10 \\ &= 0.0825. \end{aligned}$$